



Migraines

By [**Stephen D. Silberstein**](#), MD, Sidney Kimmel Medical College at Thomas Jefferson University
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A migraine headache is typically a pulsating or throbbing pain that ranges from moderate to severe. It can affect one or both sides of the head. It is often worsened by physical activity, light, sounds, or odors and accompanied by nausea, vomiting, and sensitivity to sounds, light, and/or odors.

- Migraines may be triggered by lack of sleep, changes in the weather, hunger, excessive stimulation of the senses, stress, or other factors.
- They can be made worse by physical activity, light, sounds, or odors.
- Doctors base the diagnosis on typical symptoms.
- There is no cure for migraines, but medications are used to stop the migraine as it is starting, to relieve pain, and to reduce the number and severity of migraines attacks.

(See also [Overview of Headache](#).)

Migraines are the most common cause of recurring moderate to severe headaches.

Although migraines can start at any age, they usually begin during puberty or young adulthood. In most people, migraines recur periodically (fewer than 15 days a month). After age 50, headaches often become significantly less severe or resolve entirely. Migraines are 3 times more common among women. In the United States, about 18% of women and 6% of men have a migraine at some time each year.

Migraines may become chronic. That is, they occur 15 or more days a month. Chronic migraines often develop in people who overuse medications to treat migraines.

Migraines tend to run in families. More than half the people who have migraines have close relatives who also have them.

Causes of Migraine

Migraines occur in people whose nervous system is unusually sensitive to certain stimuli. In these people, nerve cells in the brain are easily stimulated, producing electrical activity. As electrical activity

spreads over the brain, various functions, such as vision, sensation, balance, muscle coordination, and speech, are temporarily disturbed. These disturbances cause the symptoms that sometimes occur before the headache (called the aura). The headache occurs when the 5th cranial (trigeminal) nerve is stimulated. This nerve sends impulses (including pain impulses) from the eyes, scalp, forehead, upper eyelids, mouth, and jaw to the brain. When stimulated, the nerve may release substances that cause painful inflammation in the blood vessels of the brain (cerebral blood vessels) and the layers of tissues that cover the brain (meninges). The inflammation accounts for the throbbing headache, nausea, vomiting, and sensitivity to light and sound.

Estrogen, the main female hormone, appears to trigger migraines, possibly explaining why migraines are more common among women. Migraines can probably be triggered when estrogen levels increase or fluctuate. During puberty (when estrogen levels increase), migraines become much more common among girls than among boys. Some women have migraines just before, during, or just after menstrual periods. Migraines often occur less often and become less severe in the last trimester of pregnancy when estrogen levels are relatively stable, and they worsen after childbirth when estrogen levels decrease rapidly. As menopause approaches (when estrogen levels are fluctuating), migraines become particularly difficult to control.

Oral contraceptives (which contain estrogen) and estrogen therapy may make migraines worse and may increase the risk of stroke in women who have migraines with an aura.

Other triggers include the following:

- Lack of sleep, including insomnia
- Changes in the weather, particularly barometric pressure
- Red wine
- Certain foods
- Hunger (as when meals are skipped)
- Excessive stimulation of the senses (for example, by flashing lights or strong odors)
- Stress

Various foods have been associated with migraines, but whether they trigger migraines is unclear. These foods include

- Foods that contain tyramine, such as aged cheeses, soy products, fava beans, hard sausages, smoked or dried fish, and some nuts
- Foods that contain nitrates, such as hot dogs and lunch meats
- Foods that contain MSG (monosodium glutamate), a flavor enhancer found in fast foods, broths, seasonings, and spices
- Caffeine (including that in chocolate)

Specific food triggers vary from person to person.

Head injuries, neck pain, or a problem with the joint of the jaw (temporomandibular joint disorder) sometimes triggers or worsens migraines.

Familial hemiplegic migraine, a rare subtype of migraine, causes weakness on one side of the body. It is associated with genetic defects on chromosome 1, 2, or 19. The role of genes in the more common forms of migraine is under study.

In some families, family members have different migraine symptoms. Some have mainly headache. Some have mainly [vertigo](#) (a type of dizziness) or weakness on one side of the body. Some have only a migraine aura with no headache. These findings suggest that migraine may be more than just a headache disorder.

Migraine with brain stem aura is another rare subtype of migraine. It can cause [vertigo](#), [difficulty controlling and coordinating body movements](#) (ataxia), changes in vision, [tinnitus](#), slurred speech, and [fluctuations in the level of alertness](#).

Symptoms of Migraine

In a migraine, pulsating or throbbing pain is usually felt on one side of the head, but it may occur on both sides. The pain may be moderate but is often severe and incapacitating. Physical activity, bright light, loud noises, and certain odors may make the headache worse. This increased sensitivity makes many people retreat to a dark, quiet room, lie down, and sleep if possible. Typically, migraines subside during sleep.

The headache is frequently accompanied by nausea, sometimes with vomiting, and sensitivity to light, sounds, and/or odors. People have difficulty concentrating during an attack.

Attacks vary greatly in frequency and severity. Many people have several types of headache, including mild attacks without nausea or sensitivity to light. These attacks may resemble a [tension-type headache](#) but are a mild form of migraine.

Migraine attacks may last for hours to a few days (typically 4 hours to several days). Severe attacks can be incapacitating and disrupt family and work life.

A **prodrome** often occurs before a migraine. The prodrome is sensations that warn people that an attack is about to begin. These sensations may include mood changes, neck pain, food cravings, loss of appetite, and nausea.

An **aura** precedes migraines in about 25% of people. The aura involves temporary, reversible disturbances in vision, sensation, balance, muscle coordination, or speech. People may see jagged, shimmering, or flashing lights or develop a blind spot with flickering edges. Less commonly, people experience tingling sensations, loss of balance, weakness in an arm or a leg, or difficulty talking. The aura lasts minutes to an hour before and may continue after the headache begins. Some people experience an aura but have only a mild or no headache.

Migraines usually become less severe as people age. However, auras that affect vision without a headache (sometimes called ocular migraine) occur more frequently in older adults.

Did You Know...

- Sometimes migraines cause symptoms such as disturbances in vision or balance without causing a headache.
- Taking pain relievers too often can make migraines worse.

Diagnosis of Migraine

- A doctor's evaluation
- Sometimes computed tomography or magnetic resonance imaging

Doctors diagnose migraines when symptoms are typical and results of a physical examination (which includes a [neurologic examination](#)) are normal.

No test or procedure can confirm the diagnosis.

Certain findings are warning signs that suggest that the headaches may be caused by a serious disorder. These findings include the following:

- A sudden headache that becomes most severe within a few seconds or less (thunderclap headache)
- Headaches that begin after age 50
- Headaches that increase in intensity or frequency for weeks or longer
- Headaches that occur in people who have had cancer or have a weakened immune system (due to a disorder or medication)
- A severe headache accompanied by a fever, a stiff neck, and/or confusion
- Persistent problems that suggest a brain disorder, such abnormalities in sensation or vision, weakness, loss of coordination, and drowsiness or confusion
- A clear change in the established headache pattern

If headaches have developed recently or if certain warning signs are present, magnetic resonance imaging (MRI) of the head is often done, and a spinal tap (lumbar puncture) is sometimes done to exclude other disorders.

If people who are known to have migraines develop a headache that is similar to their previous migraines, doctors rarely do tests. However, if the headache is different, particularly if warning signs are present, a doctor's examination and often tests are needed.

Treatment of Migraine

- Behavioral interventions, yoga, or relaxation techniques

- Medications to stop a migraine as it starts or keep it from progressing
- Medications to control pain and nausea
- Medications to prevent migraines

Migraines cannot be cured, but they can be controlled.

Doctors encourage people to keep a headache diary. In it, people write down the number and timing of attacks, possible triggers, and their response to treatment. With this information, triggers may be identified and eliminated when possible. Then, people can participate in their treatment by avoiding triggers, and doctors can better plan and adjust treatment. However, elimination of all possible triggers can be excessive and lead to unnecessary life restrictions.

Doctors also recommend using behavioral interventions (such as relaxation, biofeedback, and stress management) to control migraine attacks, especially when stress is a trigger or when people are taking too many medications to control the migraines. Relaxation techniques can help control stress, ease muscle tension, and alter brain wave activity.

Yoga can reduce the intensity and frequency of migraines. Yoga combines physical poses that strengthen and stretch muscles with deep breathing, meditation, and relaxation.

Medications

Some medications stop (abort) a migraine as it is starting or keep it from progressing. Some are taken to control the pain. Others are taken to prevent migraines.

For mild to moderate migraines, pain relievers ([analgesics](#)) can help control the pain. Often, nonsteroidal anti-inflammatory drugs (NSAIDs) or [acetaminophen](#) is used. They can be taken as needed during a migraine, with or instead of a triptan. For occasional mild migraines, analgesics that contain [caffeine](#), an opioid, or butalbital (a barbiturate) may help. However, overuse of analgesics, [caffeine](#) (in analgesic preparations or in caffeinated beverages), or triptans can lead to daily, more severe migraines. Such headaches, called [medication overuse headaches](#), occur when these medications are taken more than 15 days a month for more than 3 months.

When migraines are or become severe, medications that can abort the migraine are used. They are taken as soon as people sense a migraine is starting. They include the following:

- **Triptans** (5-hydroxytryptamine [5-HT], or serotonin, agonists) are usually used. Triptans prevent nerves from releasing substances that can trigger migraines. Triptans are most effective when taken as soon as the migraine begins. They may be taken by mouth or by nasal spray or be injected under the skin (subcutaneously).
- **Ditans** are a class of medications that can abort headaches. They work like triptans but may have fewer side effects that involve the heart (such as changes in blood pressure or heart rate). [Lasmiditan](#), taken by mouth, is currently the only ditan being used. No more than 1 dose of [lasmiditan](#) should be taken in 24 hours.
- **Gepants** are another class of medications that can abort headaches. Gepants block a protein in the blood that triggers migraine. These medications ([rimegepant](#) and

[ubrogepant](#)) are taken by mouth.

- **Dihydroergotamine** is given intravenously, subcutaneously, and by nasal spray to stop severe, persistent migraines. It is usually given with a medication used to relieve nausea (antiemetic medication), such as [prochlorperazine](#), given intravenously.
- **Certain antiemetic medications** (such as [prochlorperazine](#) or [metoclopramide](#)) may be used to relieve mild to moderate migraines. [Prochlorperazine](#), taken by mouth or given as a suppository, is also used to stop migraines when people cannot tolerate triptans or [dihydroergotamine](#).

Because triptans and [dihydroergotamine](#) may cause blood vessels to narrow (constrict), they are not recommended for people who have angina, coronary artery disease, or uncontrolled high blood pressure. If older adults or people with risk factors for coronary artery disease need to take these medications, they must be monitored closely. However, people with one of these disorders can take [lasmiditan](#), [rimegepant](#), or [ubrogepant](#).

If migraines are usually accompanied by nausea, taking an antiemetic with a triptan when symptoms begin is effective. Antiemetics (such as [prochlorperazine](#) or [metoclopramide](#)), taken alone, may also stop mild or moderate migraines from progressing.

When other treatments are ineffective in people with severe migraines, opioid analgesics may be used as a last resort.

When migraines are severe, fluids given intravenously can help relieve headache and make people feel better, especially if people are dehydrated from vomiting.

Other medications are used to prevent migraine and can decrease the frequency and severity of symptoms. They include the following:

- Antiseizure medications
- Beta-blockers
- Calcium channel blockers
- Monoclonal antibodies ([erenumab](#), [fremanezumab](#), and [galcanezumab](#))
- Tricyclic antidepressants

[OnabotulinumtoxinA](#) (which is injected into the scalp, forehead, and neck) or monoclonal antibodies can be used to treat chronic migraines.

Missing or reducing a dose of a medication used to prevent migraines or taking it late may trigger or worsen a migraine.

TABLE

Some Medications Used to Treat Migraines

Type

Examples

Some Side Effects

Prevention

Type	Examples	Some Side Effects
Antiseizure medications	Divalproex Valproate	Hair loss, stomach upset, liver dysfunction, a tendency to bleed, tremors, and weight gain Not used in pregnant women with migraines
	Topiramate	Weight loss, confusion, and depression
Beta-blockers	Atenolol Metoprolol Nadolol Propranolol Timolol	Spasm of the airways (bronchospasm), an abnormally slow heart rate (bradycardia), fatigue, insomnia, worsening of heart failure, and sexual dysfunction Changes in blood sugar levels (possibly making it difficult for doctors to know when blood sugar levels become too low in people with diabetes) With some beta-blockers, unfavorable effects on lipid (fat) levels
A botulinum toxin (used to block nerve activity)	OnabotulinumtoxinA	Neck pain and stiffness Weakness of muscles in the face and sometimes the neck
Calcium channel blockers	Verapamil	Dizziness, low blood pressure, and weakness Constipation
Ditans*	Lasmiditan	Dizziness and drowsiness
Monoclonal antibodies	Erenumab Fremanezumab Galcanezumab	Pain and irritation at the injection site and constipation

Type	Examples	Some Side Effects
		Very rarely, allergic reactions
Tricyclic antidepressants	Amitriptyline	Drowsiness, weight gain, increased heart rate, dry mouth, confusion, and constipation
Treatment of severe migraines		
Antiemetic medications (used to relieve nausea)	Metoclopramide Prochlorperazine	Low blood pressure, drowsiness, involuntary movements, and muscle spasms
Ergot derivatives	Dihydroergotamine	Medication overuse headache (rebound headache) if the medication is taken too frequently Nausea, vomiting, and minor muscle cramping Rarely, chest pain due to an inadequate blood supply to the heart muscle (angina)
Gepants*	Rimegepant Ubrogepant	Minimal
Triptans (5-hydroxytryptamine [5-HT], or serotonin, agonists)	Almotriptan Eletriptan Frovatriptan Naratriptan Rizatriptan Sumatriptan Zolmitriptan	Medication overuse headache if the medication is taken too frequently Flushing, tingling, dizziness, drowsiness, nausea, and a sense of pressure or pain in the throat or chest
Opioids †	Codeine Meperidine	Medication overuse headache if the

Type	Examples	Some Side Effects
	<u>Oxycodone</u>	medication is taken too frequently Slowing of breathing, constipation, retention of urine, drowsiness, and nausea
Treatment of mild to moderate migraines		
Analgesics	<u>Acetaminophen</u>	Medication overuse headache if the medication is taken too frequently Occasionally rash
Antiemetic medications	<u>Metoclopramide</u> <u>Prochlorperazine</u>	Low blood pressure, drowsiness, involuntary movements, and muscle spasms
Nonsteroidal anti-inflammatory drugs (<u>NSAIDs</u>)	<u>Aspirin</u> <u>Ibuprofen</u> <u>Indomethacin</u> <u>Ketorolac</u> <u>Naproxen</u>	Medication overuse headache if the medication is taken too frequently Gastritis (inflammation of the stomach) and peptic ulcers With <u>indomethacin</u>, worsening of depression, seizures, and tremors with decreased mobility and muscle stiffness and, in older adults, dizziness and confusion If <u>ketorolac</u> is overused, possibly impaired kidney function

* Ditans can also be used to treat mild to moderate and severe migraines.

† Opioids should be used as a last resort when migraines cause severe pain and other measures are ineffective.

Devices that stimulate certain nerves, held to certain parts of the body, can help treat migraine attacks, as well as prevent them. One such device, held to the forehead, can reduce the frequency of migraines. Another device, held to the back of the skull, can relieve migraines as they start. A device that uses an armband to stimulate nerves in the skin can relieve migraine pain. These devices have no significant side effects.

Prevention of Migraine

When treatment does not prevent people from having frequent or incapacitating migraines, taking medications every day to prevent migraine attacks can help (see table [Some Medications Used to Treat Migraines](#)). Taking preventive medications may decrease the need for pain relievers or other migraine medications and thus help avoid [medication overuse headache](#).

The choice of a preventive medication is based on the side effects of the medication and on other disorders present, as in the following examples:

- [Beta-blockers](#), such as [propranolol](#), are often used, particularly in people with anxiety or coronary artery disease.
- The antiseizure medication [topiramate](#) may be given to people who are overweight because it can promote weight loss.
- The antiseizure medication divalproex can help stabilize mood and may be useful if migraines make functioning difficult.
- [Amitriptyline](#) may be given to people with depression or insomnia.
- [OnabotulinumtoxinA](#) (used to block nerve activity) or certain other medications (such as divalproex and monoclonal antibodies) may be used when other medications are ineffective.
- Monoclonal antibodies (such as [erenumab](#), [fremanezumab](#), or [galcanezumab](#)) if other medications are ineffective.
- Gepants, such as [atogepant](#) or [rimegepant](#), can be used for migraine prevention.

Monoclonal antibodies, given by injection, block the action of a substance that can trigger migraines.

Other medications that can be used to prevent migraines include calcium channel blockers.

Devices that stimulate certain nerves, held to certain parts of the body, can also help prevent migraine attacks. One such device, held to the forehead, can reduce the frequency of migraines. Another device,

held to the back of the skull, can relieve migraines as they start and help prevent them. These devices have no significant side effects.

Drug Information for the Topic



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